

2c. Content of Computer systems (Component 01)

This component will introduce learners to the internal workings of the Central Processing Unit (CPU), the exchange of data and will also look at software development, data types and legal and ethical issues. It is expected that learners will draw on this underpinning content when studying computational thinking, developing programming techniques and

devising their own programming approach in the Programming project component (03 or 04).

Learners will be expected to apply the criteria below in different contexts including current and future uses of the technologies.

1.1 The characteristics of contemporary processors, input, output and storage devices	
Components of a computer and their uses	
1.1.1 Structure and function of the processor	(a) The Arithmetic and Logic Unit; ALU, Control Unit and Registers (Program Counter; PC, Accumulator; ACC, Memory Address Register; MAR, Memory Data Register; MDR, Current Instruction Register; CIR). Buses: data, address and control: how this relates to assembly language programs. (b) The Fetch-Decode-Execute Cycle; including its effects on registers. (c) The factors affecting the performance of the CPU: clock speed, number of cores, cache. (d) The use of pipelining in a processor to improve efficiency. (e) Von Neumann, Harvard and contemporary processor architecture.
1.1.2 Types of processor	(a) The differences between and uses of CISC and RISC processors. (b) GPUs and their uses (including those not related to graphics). (c) Multicore and Parallel systems.
1.1.3 Input, output and storage	(a) How different input, output and storage devices can be applied to the solution of different problems. (b) The uses of magnetic, flash and optical storage devices. (c) RAM and ROM. (d) Virtual storage.